

ORF309 Midterm 1 Q1c: Understanding the Common Errors

The Question

Let N be the number of times during one week that you eat banana yogurt **but do not get a rash**. Find $\mathbb{E}[N]$.

What You Did (and Why It's Problematic)

You calculated $n \times p = 7 \times 0.35 = 2.45$ and got the **right answer**.

However, this suggests you were thinking of N as a **Binomial distribution** with parameters $(n = 7, p = 0.35)$, where:

- $n = 7$ trials (days)
- $p = 0.35$ is the probability of “success” (eating banana yogurt and not getting a rash) on each day

The Problem

Binomial distributions require independence between trials.

The question **never states** that the days are independent. In fact, it's reasonable to think the yogurt choices might be dependent (e.g., they might run out of banana yogurt if someone takes it all on Monday).

The Correct Approach: Linearity of Expectation

Step 1: Define Indicator Variables

For each day $i \in \{1, 2, \dots, 7\}$, define:

$$N_i = \begin{cases} 1 & \text{if you eat banana yogurt AND don't get a rash on day } i \\ 0 & \text{otherwise} \end{cases}$$

Each N_i is a **Bernoulli random variable** with parameter $p = 0.35$.

Step 2: Express N as a Sum

The total number of times you eat banana yogurt without getting a rash in the week is:

$$N = \sum_{i=1}^7 N_i$$

Step 3: Apply Linearity of Expectation

Key Insight

Linearity of expectation states:

$$\mathbb{E} \left[\sum_{i=1}^7 N_i \right] = \sum_{i=1}^7 \mathbb{E}[N_i]$$

This does NOT require independence!

Since each N_i is a Bernoulli random variable with $p = 0.35$:

$$\mathbb{E}[N_i] = p = P(\text{eat banana yogurt AND no rash on day } i)$$

Step 4: Calculate the Probability p

From the problem:

$$\begin{aligned} p &= P(\text{eat banana yogurt AND no rash}) \\ &= P(\text{no rash} \mid \text{banana yogurt}) \times P(\text{banana yogurt}) \\ &= P(NR \mid E_2) \times P(E_2) \\ &= 0.7 \times 0.5 \\ &= 0.35 \end{aligned}$$

Step 5: Compute the Final Answer

$$\mathbb{E}[N] = \sum_{i=1}^7 \mathbb{E}[N_i] = \sum_{i=1}^7 0.35 = 7 \times 0.35 = 2.45$$

Why Your Answer is Right but Your Reasoning is Wrong

Your Approach (Binomial)	Correct Approach (Linearity)
$N \sim \text{Binomial}(7, 0.35)$	Define $N = \sum_{i=1}^7 N_i$
$\mathbb{E}[N] = n \times p = 7 \times 0.35$	$\mathbb{E}[N] = \sum_{i=1}^7 \mathbb{E}[N_i] = 7 \times 0.35$
Requires independence	No independence needed
Not justified by the problem	Valid reasoning

Summary

1. Your calculation $7 \times 0.35 = 2.45$ is **numerically correct**.
2. However, if you assumed $N \sim \text{Binomial}(7, 0.35)$, you implicitly assumed **independence**, which is **not stated in the problem**.
3. The **correct justification** is **linearity of expectation**, which works **without requiring independence**.

4. This is a subtle but important distinction: **the formula $\mathbb{E}[N] = np$ for binomial distributions happens to equal the linearity of expectation result, but the reasoning matters!**

Takeaway

Always use linearity of expectation when calculating $\mathbb{E}[\text{sum of indicators}]$, because it doesn't require independence.

Only invoke binomial distributions when you can explicitly justify independence.